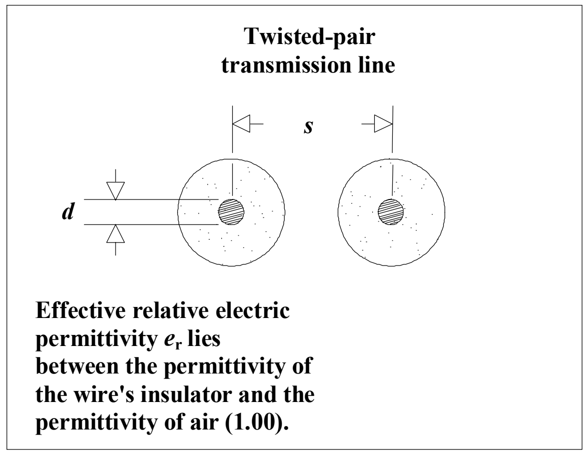
TRANSMISSION LINE MADE FROM TWISTED PAIR WIRE file: twist.mcd

Formulas included in this spreadsheet:

Twisted-pair characteristic impedance	ZTWIST()
Twisted-pair propagation delay	PTWIST()
Twisted-pair inductance	LTWIST()
Twisted-pair capacitance	CTWIST()

Variables used:

- d Diameter of wire (in.)
- s Separation between wires (in.)
- x Length of wire (in.)
- er Effective relative dielectric constant of medium between wires



twist

Characteristic impedance of twisted pair (Ω):

$$\mathrm{ZTWIST}(d,s,er) := \frac{120}{\sqrt{er}}\cdot\ln\left(\frac{2\cdot s}{d}\right)$$

Propagation delay per in. twisted pair (s/in.):

$$\mathrm{PTWIST}(er) := 84.72\cdot 10^{-12}\cdot\sqrt{er}$$

Inductance of twisted pair (H):

$$\mathrm{LTWIST}(d,s,x) := x\cdot 10.16\cdot 10^{-9}\cdot\ln\left(\frac{2\cdot s}{d}\right)$$

Capacitance of twisted pair (F):

$$\mathrm{CTWIST}(d,s,er,x) := \left(\frac{x\cdot .7065\cdot 10^{-12}}{\ln\left(\frac{2\cdot s}{d}\right)}\right)\cdot er$$

Example twisted-pair calculations

Diameter of AWG 24 wire (in.)	$D := .02$
Length of wire (in.)	$X := 2.000$
Separation between wire centers (in.)	$S := .038$
Relative dielectric constant	$er := 2.5$

Characteristic impedance (Ω):

$$\mathrm{ZTWIST}(D, S, er) = 101.319$$

Total inductance (H):

$$\mathrm{LTWIST}(D, S, X) = 2.713 \times 10^{-8}$$

Same result in nH:

$$\mathrm{LTWIST}(D, S, X) \cdot 10^{9} = 27.127$$

Inductance per in. (H):

$$\mathrm{LTWIST}(D, S, 1) = 1.356 \times 10^{-8}$$

Total capacitance (F):

$$\mathrm{CTWIST}(D, S, er, X) = 2.646 \times 10^{-12}$$

Same result in pF:

$$\mathrm{CTWIST}(D, S, er, X) \cdot 10^{12} = 2.646$$

Capacitance per in. (F):

$$\mathrm{CTWIST}(D, S, er, 1) = 1.323 \times 10^{-12}$$